CS 1671 Progress Report - Group 10 (Amanda Cotumaccio, Jay Patel, Jonah Belback, Lyndsey Dippold, and Saung Yati Oo)

1. **Basic data analysis:** Information about dataset
    1. If it has been updated from the proposal, provide the number of rows (datapoints) and columns in the dataset and what each datapoint and column corresponds to. If you are splitting the dataset into a training, test, and possible dev sets, how many rows are in each?

Instead of using 641,349 output labels and the whole dataset we initially proposed, we plan to use only the top 20 artists. Each of these artists have over 100 songs each, 4525 total, however there is a large distribution between them. We plan on fixing this by making a subset of the total 4525-point dataset made up of 100 randomly selected songs from each artist (total of 2000 songs), and then redistribute this subset into 70% training, 15% validation, and 15% evaluation. This will ensure a more even distribution without the model overfitting to artists with more songs than others, and prevent requiring a high level of complexity to fit a huge number of labels.

Columns per datapoint:

1. Title – Title of the piece
2. Tag – Genre classification (e.g., pop, rap)
3. Artist – Author/artist of the piece (target label)
4. Year – Year released
5. Views – Number of page views
6. Features – Featured artists
7. Lyrics – Full multiline string that contains the main text of the piece.
8. Id - unique identifier
9. Language_cld3 - lyrics language according to the cld3 model
10. Language_ft - lyrics language according to the FastText language identification model
11. Language: - Propagates the value of language_cld3 and language_ft if they are equal, otherwise is NaN

Number of datapoints (rows) in total dataset of top 20 artists: 4525
Number of datapoints (rows) in subset: 2000
Number of datapoints (rows) in training subsubset: 1400
Number of datapoints (rows) in validation subsubset: 300
Number of datapoints (rows) in evaluation subsubset: 300

We will only use the lyrics column as input features, and the artist column as the target label. All other columns are not used in the model.

2. Optionally, any other distribution or data visualization that you think is helpful for understanding your dataset or task.

Here is the output of our dataset using the top 20 artists which are listed here with their amount of songs. Below this table is an example of a datapoint.

```
Columns:
Index(['title', 'tag', 'artist', 'year', 'views', 'features', 'lyrics', 'id',
       'language_cld3', 'language_ft', 'language'],
      dtype='object')

Top 20 artists:
artist
Barbra Streisand         446
Andr Hazes               280
The Beach Boys           263
Chris De Burgh           238
Madonna                  233
Franco Battiato          231
Sezen Aksu               229
Miguel Bos               226
Roy Orbison              218
Steven Curtis Chapman    218
Kylie Minogue            215
Michael Jackson          205
Mariah Carey             199
New Model Army           197
Alceu Valena             193
Alkehol                  191
Acappella                191
Franco Califano          189
Christy Moore            184
Lana Del Rey             179
Name: count, dtype: int64
title                                              The Way We Were
tag                                                            pop
artist                                           Barbra Streisand
year                                                         1973
views                                                       86182
features                                                       {}
lyrics           [Intro]\nHmmmmm hmmmmm\nHmm hmm hmmmmmm hmmm h...
id                                                         632449
language_cld3                                                  en
language_ft                                                    en
language                                                       en
Name: 395423, dtype: object

Final dataset shape: (4525, 11)
```

2. **A result from baseline (traditional) approach:** In your proposal, you described an initial baseline approach to your task, which for most groups was using n-gram features in some way. Please provide one (hopefully quantitative) result from your work so far in this direction. Ideally this would be a performance metric result from your baseline approach on a dev or test set. But if you're not that far yet, you can also provide an example of working input and output from your system or part of a system, some sort of plot or other output. You can be up front about challenges you are facing for which you might need help; to get a good grade, I'll just be looking for some sort of output from a working system or part of a system. If you are confused what this means for your project, contact the instructor.

```
Alkehol
Amanda1@Amandas-MacBook-Pro-3 PitchPerfect-main % python3 baseline_lyrics_classifier.py
Dataset shape: (4525, 11)

Artists:
artist
Barbra Streisand         446
Andr Hazes               280
The Beach Boys           263
Chris De Burgh           238
Madonna                  233
Franco Battiato          231
Sezen Aksu               229
Miguel Bos               226
Steven Curtis Chapman    218
Roy Orbison              218
Kylie Minogue            215
Michael Jackson          205
Mariah Carey             199
New Model Army           197
Alceu Valena             193
Alkehol                  191
Acappella                191
Franco Califano          189
Christy Moore            184
Lana Del Rey             179
Name: count, dtype: int64

Train size: 3620
Dev size: 905

MODEL 1: UNIGRAM BOW
Accuracy: 0.6895027624309392
                        precision    recall  f1-score   support

            Acappella       0.85      0.76      0.81        38
         Alceu Valena       0.97      1.00      0.99        39
              Alkehol       0.97      1.00      0.99        38
           Andr Hazes       0.98      0.93      0.95        56
     Barbra Streisand       0.52      0.66      0.58        89
       Chris De Burgh       0.59      0.64      0.61        47
        Christy Moore       0.61      0.68      0.64        37
      Franco Battiato       0.67      0.80      0.73        46
      Franco Califano       0.75      0.79      0.77        38
        Kylie Minogue       0.41      0.33      0.36        43
         Lana Del Rey       0.59      0.61      0.60        36
             Madonna        0.60      0.53      0.56        47
        Mariah Carey        0.56      0.47      0.51        40
      Michael Jackson       0.60      0.37      0.45        41
           Miguel Bos       0.89      0.73      0.80        45
       New Model Army       0.81      0.77      0.79        39
          Roy Orbison       0.40      0.48      0.44        44
           Sezen Aksu       0.98      1.00      0.99        46
Steven Curtis Chapman       0.72      0.64      0.67        44
       The Beach Boys       0.58      0.62      0.60        52

             accuracy                           0.69       905
            macro avg       0.70      0.69      0.69       905
         weighted avg       0.69      0.69      0.69       905


MODEL 2: TF-IDF
Accuracy: 0.6861878453038674
                        precision    recall  f1-score   support

            Acappella       0.79      0.68      0.73        38
         Alceu Valena       0.97      1.00      0.99        39
              Alkehol       0.97      1.00      0.99        38
           Andr Hazes       1.00      0.95      0.97        56
     Barbra Streisand       0.42      0.76      0.55        89
       Chris De Burgh       0.68      0.55      0.61        47
        Christy Moore       0.73      0.65      0.69        37
      Franco Battiato       0.80      0.78      0.79        46
      Franco Califano       0.76      0.89      0.82        38
        Kylie Minogue       0.36      0.35      0.35        43
         Lana Del Rey       0.64      0.58      0.61        36
             Madonna        0.52      0.49      0.51        47
        Mariah Carey        0.64      0.45      0.53        40
      Michael Jackson       0.56      0.34      0.42        41
           Miguel Bos       0.85      0.73      0.79        45
       New Model Army       0.69      0.87      0.77        39
          Roy Orbison       0.57      0.30      0.39        44
           Sezen Aksu       1.00      1.00      1.00        46
Steven Curtis Chapman       0.76      0.59      0.67        44
       The Beach Boys       0.61      0.65      0.63        52

             accuracy                           0.69       905
            macro avg       0.72      0.68      0.69       905
         weighted avg       0.70      0.69      0.68       905
```

```
SAMPLE INPUT (first 500 characters):
Když se blíží léta čas
Začne to vřít v každym z nás
Už se všichni těšíme
Jak ten příchod oslavíme

R: Léto už je tady ukazuje svoje vnady
Dokud ještě žijeme v létě rádi pijeme
Orosený lahváče ať abstinent zapláče
No bóže co je natom – hlavně že nám chutná

Dveře zahrad se votvíraj
Tyhle finty na nás zabíraj
Už je slyšet pípy zpěv
A my pějem chvalozpěv

R: Léto už je tady ukazuje svoje vnady
Dokud ještě žijeme v létě rádi pijeme
Orosený lahváče ať abstinent zapláče
No bóže co je natom – hlavně že

TRUE ARTIST:
Alkehol

PREDICTED ARTIST (TF-IDF):
Alkehol
```

For our baseline (traditional) approach, we adapted our HW2 text classification pipeline to predict the artist from song lyrics. We used n-gram-based features, specifically unigram bag-of-words and TF-IDF representations, with logistic regression.

We filtered the dataset to include only pop songs from the top 20 most frequent artists, resulting in 4,525 samples. The data was split into 3,620 training examples and 905 development examples.

Our unigram bag-of-words model achieved an accuracy of 68.95% on the development set, while the TF-IDF model achieved 68.62% accuracy. We also verified that the system works end-to-end by providing a sample lyric input and confirming that the model outputs a predicted artist label (shown in the figure above).

3. **LLM proposal:** In the project, you will be comparing your baseline system's performance to that of an LLM. Please describe how you might use an LLM programmatically to attempt your task. The simplest way to do this would be in a "zero-shot" setting where you simply ask the LLM to do the task, but even that requires setting up and passing your data to the LLM and evaluating it. Please describe what you plan to do and which LLM you plan on using. You can also propose using more advanced approaches such as in-context learning (few-shot prompting), chain-of-thought prompting, prompt optimization or fine-tuning. Not all groups have to use an LLM here if you have already talked to the instructor; in that case, please describe the rest of the approach you will be taking to complete the project.

   We plan on comparing the performance of our model to an LLM by making a second model using BERT[1] to encode input text into a vector followed by a simple model (either linear layer or MLP) and reuse the same classification header as our model. We don't see this as being too much to accomplish as one of our members is familiar with this process and has preexisting resources. If this approach is not enough, we can instead use cosine similarity from the BERT encoded vectors instead of a simple model.

   If BERT doesn't fit this LLM requirement, we can alternatively use the encoder part of a transformer[2] model.

   This comparison will use the same metrics and dataset between the BERT model and our own proposed N-gram model.

   Another approach would be to evaluate a large language model (LLM) using API access. We would prompt the model with song lyrics and ask it to predict the artist from our fixed set of 20 artists. For example, we may provide a prompt that includes the list of possible artists and the lyrics, and instruct the model to output a single artist name.

   For few-shot prompting, we may include a small number of labeled examples in the prompt to guide the model's predictions. We will evaluate the LLM using the same dataset and metrics as our other models (accuracy, precision, recall, and F1 score) to ensure a fair comparison.

[1] Devlin, Jacob, et al. "BERT: Pre-Training of Deep Bidirectional Transformers for Language Understanding." *ArXiv*, 11 Oct. 2018, arxiv.org/abs/1810.04805.
[2] Vaswani, Ashish, et al. "Attention Is All You Need." *Cornell University*, 12 June 2017, arxiv.org/abs/1706.03762.

4. **Open questions and challenges:** Please describe any open questions or challenges your group has at this point. Will you need any resources other than the ones provided in class (OpenAI API access, CRCD access) or have any other questions? Also describe if the roles for each of your team members have changed since the proposal and if so, what the new roles are.

   We will require CRCD access for processing and training our models, particularly for handling the large dataset and running experiments efficiently.

   One challenge we anticipate is handling the length of song lyrics, as some models (especially LLMs) have input token limits. We may need to truncate or preprocess lyrics, which could affect performance.

   Another challenge is ensuring fair evaluation across models, since the dataset is originally imbalanced and requires careful sampling to avoid bias.

   The only change in roles:
   Jonah Belback will do the LLM performance comparison, including

   - Making any LLM models
   - Developing/Training any LLM models
   - Evaluation of any LLM models
   - Comparison between group's model